

Property-Guided Few-Shot Learning for Molecular Property Prediction With Dual-View Encoder and Relation Graph Learning Network

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Abstract—Molecular property prediction is an important task in drug discovery. However, experimental data for many drug molecules are limited, especially for novel molecular structures or rare diseases which affect the accuracy of many deep learning methods that rely on large training datasets. To this end, we propose PG-DERN, a novel few-shot learning model for molecular property prediction. A dual-view encoder is introduced to learn a meaningful molecular representation by integrating information from node and subgraph. Next, a relation graph learning module is proposed to construct a relation graph based on the similarity between molecules, which improves the efficiency of information propagation and the accuracy of property prediction. In addition, we use a MAML-based meta-learning strategy to learn well-initialized meta-parameters. In order to guide the tuning of meta-parameters, a property-guided feature augmentation module is designed to transfer information from similar properties to the novel property to improve the comprehensiveness of the feature representation of molecules with novel property. A series of comparative experiments on four benchmark datasets demonstrate that the proposed PG-DERN outperforms state-of-the-art methods.

Index Terms—Molecular property prediction, few-shot learning, graph neural network, meta-learning.

I. INTRODUCTION

DRUG discovery involves a series of complex steps such as identification of disease targets, drug screening, design optimization, and clinical trials, which is a long, expensive and inefficient process [1]. In general, the process of drug discovery consumes \$2.8 billion and spans over 15 years approximately, while the success rate is only around 10% [2]. How to reduce costs and accelerate drug discovery has become a challenging issue.

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Code link: <https://github.com/Bombtsti/PG-DERN>.

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Molecular property prediction is significant in various phases of drug discovery, which can effectively speed up the process of drug discovery [3]. In drug screening, property prediction can predict the pharmacological activity and toxicity of molecule to narrow the scope of candidate compounds. In clinical trials, molecular property prediction can predict the potential adverse reactions, biological activities and other properties of drugs, which is helpful in assessing the potential risks of candidate drugs. Quantitative structure property/activity relationship (QSPR/QSAR) is a method to establish relationships between the structure features of a compound and its biological activity [4]. Benefiting from the development of deep learning, QSPR/QSAR can predict the molecular property more powerfully and accurately [5], thus accelerating the design and optimization of drug molecules.

When using the methods based on deep learning models, the first problem is the way of representing a molecule. Depending on the input type, the representation method can be categorized into two types: sequence and molecular graph. For the first type, molecules are encoded into the unique sequence by certain rules. The most common type is Simplified Molecular Input Line Entry Specification (SMILES), which encodes atoms and chemical bonds in a molecule as characters and form a string through specific rules. This method can typically utilize language models in natural language processing to learn the representation of molecules [6]. Jiang et al. [7] proposed a multi-granularity SMILES molecular learning model, which enhanced the molecular representation from three different granularities: node, substring, and graph. Liu et al. [8] extracted embedding from SMILES strings and used it as input to bidirectional recurrent gated (BiGRU) to learn contextual feature. However, sequence-based representations are not effective for modeling molecule because they are lack of accurate structural description of the molecule [9].

On the other hand, molecule can be depicted as a graph structure [10], with atoms as nodes and chemical bonds as edges, where each atom is connected to a neighboring atom by different types of bonds [11]. Graph neural network (GNN) can effectively capture the structure information and relationships between nodes in graph. Therefore, a large amount of GNN models have been applied to molecular property prediction [12]. Specifically, CD-MGNN [13] develops a multi-view GNN to generate graph embedding from the perspective of nodes and

edges respectively, then utilizes a cross-dependent messaging scheme to integrate information from nodes and edges, which enhances the comprehensive of the model. HMGNN [14] captures structural semantic information of molecules and integrates multi-order information into molecular representations by encoding motif structures.

In fact, molecular property is formed by the superposition of positive and negative contributions of each substructure fragment in the molecule. Some substructure fragments have a positive effect on molecular property and some have a negative effect. Therefore, the analysis of molecular substructures is crucial. In graph representation learning, we usually use a subgraph to represent a particular substructure of a molecule. However, most of the methods mentioned above use the node-based GNN, which are insufficient for subgraph analysis.

The subgraph-based GNN methods are widely applied in graph classification. They first obtain a node-centered subgraph, and then directly use GNN to calculate the representation of the subgraph to obtain more accurate local structure information. Specifically, Chen et al. [15] proposed a structure-aware transformer based on a self-attention mechanism that utilized the self-attention mechanism to focus on different subgraphs. In molecular property prediction, the local substructure information of molecules can be well captured using subgraph-based GNN. However, the information and relationships in many graph data are context-dependent. For complex types of graph data such as molecules, the relationships between nodes may span multiple substructures. Therefore, partitioning the complete graph into multiple substructures may cause these relationships not to be captured, leading to the loss of information [16]. So, compensating for the respective shortcomings of node-based GNN and subgraph-based GNN is necessary to improve the performance of the model, which is also an important issue to be solved in this paper.

In addition, the performance of machine learning models usually depends on the size and quality of the data [17]. For many molecular properties, the amount of experimental data that can be obtained is relatively small, which results in limited labeled data and seriously affects the model performance. Therefore, molecular property prediction is actually a few-shot learning problem [18]. Unfortunately, the previously mentioned approach cannot be applied to few-shot scenarios.

Few-shot learning is a machine learning method that can use limited labeled data to build an effective prediction model. It plays an important role facing tasks with sparse or expensive experimental data [19]. The current popular approach is to apply meta-learning to traditional few-shot tasks. The idea underlying meta-learning is to learn from a number of few-shot tasks to obtain a model with good discrimination and generalization ability to unknown categories [20].

Meta-learning methods can generally be divided into three categories: (1) model-based methods; (2) metric-based methods; (3) optimization-based methods. Model-based methods focus on quickly adapting model parameters with a small amount of data. MANN [21] employed an external memory network for acquiring cross-task knowledge during the process of meta-learning. Metric-based methods learn to measure the similarity of the input data for categorization. Prototypical Networks [22] uses

the average feature vectors of different categories of support data as the class prototype, then selects the category of the prototype closest to the query data as the prediction of the query data. Relation Network [23] constructs a distance metric by deep neural network to compare a small amount of input data within episodes. Optimization-based methods utilize specific meta-learner to learn well-initialized model parameters for different tasks. MAML [24] learns good initial model parameters from training tasks and adapts to a novel task after a small number of iterations or gradient updates.

For few-shot molecular property prediction, Altae-Tran et al. [25] introduced a new iteratively refined LSTM combined with GNN, which significantly improved the capacity of model to learn distance metrics for a limited set of molecules. Meta-MGNN [9] integrates self-supervised learning and attention-based task weights into meta-learning framework. PAR [26] proposes a property-aware embedding function to project the molecular embedding to the substructure-aware space, and utilizes the relationship between molecules to propagate limited labels. HSL-RG [27] uses graph kernels to construct global relationship graphs to learn molecular representations and designs self-supervised learning algorithms to capture molecular structure semantics. Meta-MGNN, PAR and HSL-RG both use MAML as a meta-learning framework. Although the goal of MAML is to adapt to a small amount of data, the limited information contained in few data, as well as the offset in their distribution compared to the ground truth distribution, will impair the generalization capability of the model. In addition, the differences between the source and target domains affect the performance of the model [28], so considering similarities between different tasks might enhance the effect of meta-learning.

Realizing the above challenges, we proposed a novel model, named PG-DERN, which is designed to address the problem of data scarcity and information deficiency in few-shot molecular property prediction through graph models and meta-learning. Specifically, our main contributions can be summarized as follows:

- A dual-view encoder module is proposed which aims to learn graph embedding from two different perspectives of node and subgraph. The two view encoders complement each other's structural information for feature augmentation. To ensure the stability of dual-view encoder, a disagreement loss is introduced to balance the difference in dual-view prediction.
- A relation graph learning module is used to construct a relation graph based on the similarities between molecules. The information of molecules can be propagated on the relation graph, allowing the molecules to learn additional supervised information from other molecules, thus strengthening the robustness of the molecule features.
- Inspired by Yang et al. [29], we propose a property-guided feature augmentation module. We calculate the similarities between the novel property and other properties and transfer the knowledge of molecules with similar properties to the novel property molecules, augmenting the feature of the novel property molecules on the support set to guide the direction of parameter optimization.

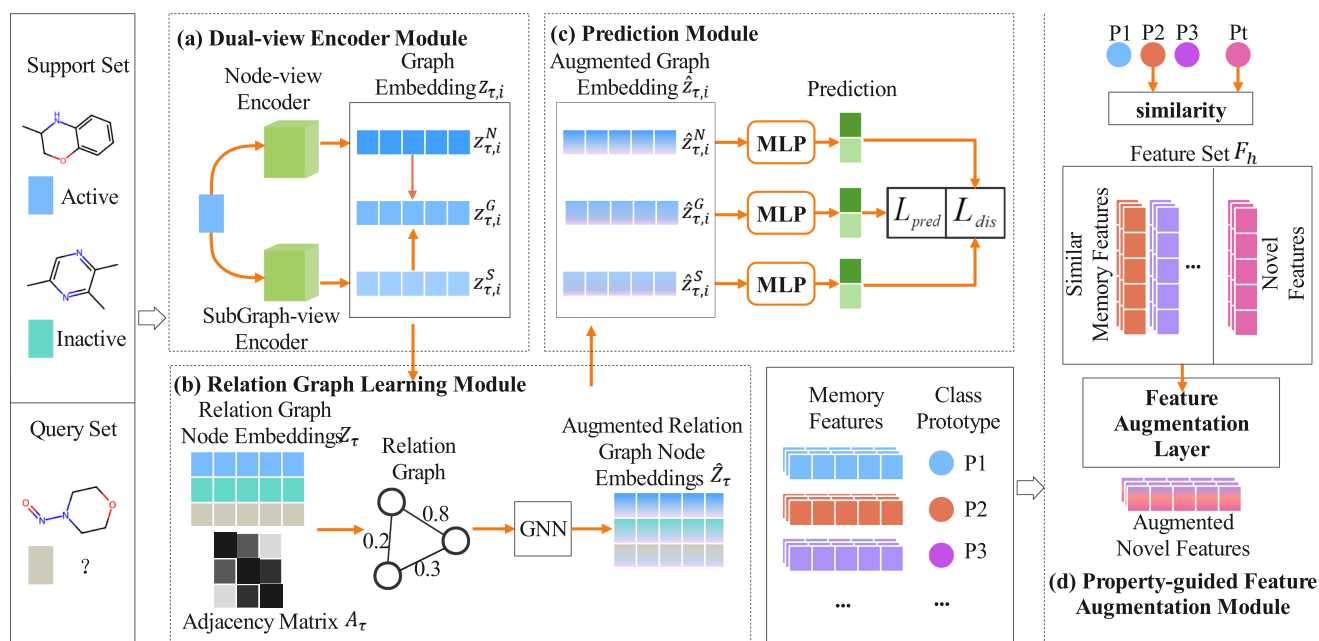


Fig. 1. Work flow of PG-DERN.

II. METHODS

A. Problem Definition

Following the problem definition proposed in Meta-MGNN [9] and PAR [26], for the given N_t few-shot molecular property prediction tasks $\{T_{\tau}\}_{\tau=1}^{N_t}$, the τ th task T_{τ} is defined as a 2-way-K-shot classification task. Each task contains support set $S_{\tau} = \{(x_{\tau,i}, y_{\tau,i})\}_{i=1}^{2K}$ with $2K$ samples and query set $Q_{\tau} = \{(x_{\tau,j}, y_{\tau,j})\}_{j=1}^{N_{\tau}^q}$ with N_{τ}^q samples, where $x_{\tau,i}$ and $y_{\tau,i} \in \{0, 1\}$ denote molecule and whether the molecule is active in the current property. Our goal is to learn a meta-learner with good initialized parameters from a set of training tasks. When given a novel task with only a small number of labeled molecules, meta-learner is able to predict the activity of unknown molecules on the novel property after fine-tuning the parameters on the support set.

B. Overview

In this paper, we propose a novel method (PG-DERN) to solve the problem of molecular property prediction in few-shot scenarios, the work flow of PG-DERN is shown in Fig. 1. First, the graph embedding of molecule in each task is obtained through a specially designed dual-view encoder module. Then, the molecular graph embedding in the current task is fed into the relation graph learning module, in which each query molecule and all support molecules are constructed as a relation graph, and the cosine similarity between each molecule is calculated as its adjacency matrix in the relation graph. With the help of the relation graph learning module, the graph embedding of molecule can be augmented by incorporating structure knowledge of other molecules. Subsequently, the augmented molecular graph embeddings are fed into the prediction module which contains three Multi-Layer Perceptions (MLP). In addition, the

disagreement loss is designed to stabilize the training procedure of the dual-view encoder.

In the meta-learning step, the MAML-based meta-learning framework is used for training. During meta-training, the active molecular embeddings corresponding to each property are stored as memory features. The class prototype is obtained by averaging the memory features and can be used as the representation of the current property. In the fine-tuning stage of the meta-testing, we design a property-guided feature augmentation module. It obtains a set of properties similar to the novel property by calculating the similarity between class prototypes, then the novel molecular features are augmented using the memory features with similar properties. Following this, meta-learner can better fine-tune the parameters to obtain the base-learner with better generalization capability, thus improving the accuracy of prediction for unknown molecules. Next, we will introduce the specific steps of each module in detail.

C. Dual-View Encoder Module

Molecule can be represented as a topological graph $G = (V, E)$, an atom can be represented as a node and a chemical bond can be represented as an edge connecting two nodes, where V is the set of atoms and E is the set of chemical bonds. Graph Neural Network (GNN) captures the information of molecular structure, atoms and chemical bonds through a message passing mechanism, which updates the node embedding by aggregating the information of neighboring node. After message passing, the graph embedding is obtained by aggregating all node embeddings and can be regarded as the feature representation of the molecule.

Node-view encoder: In this module, the graph isomorphic network (GIN) [30] is used as the node-view encoder. The GIN

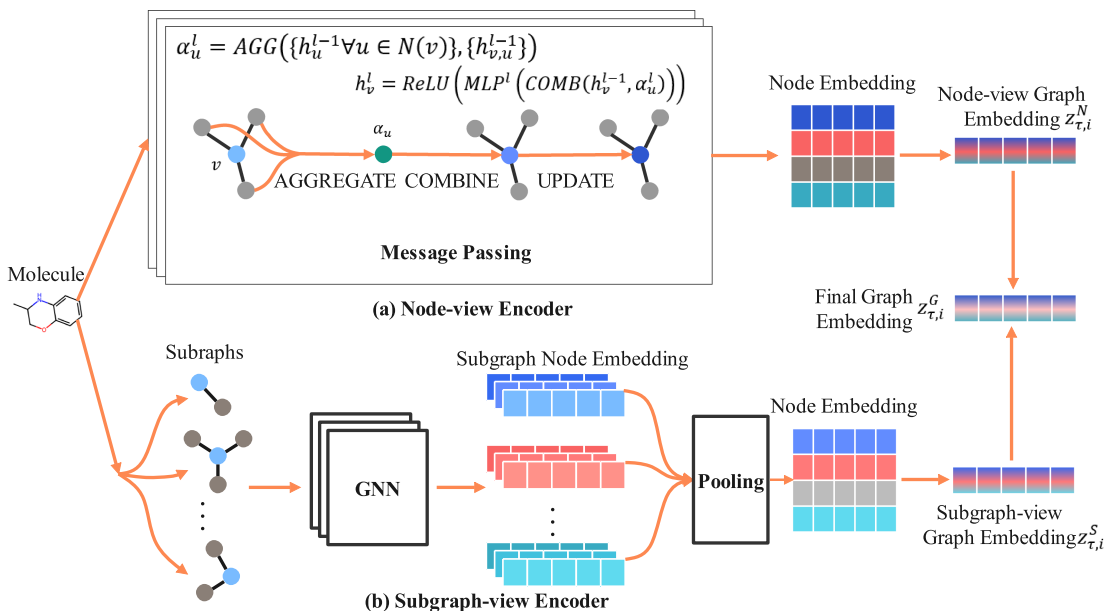


Fig. 2. Architecture of the dual-view encoder.

message passing process, as illustrated in Fig. 2(a), is executed through the *AGGREGATE* and *COMBINE* procedures. In the *AGGREGATE* step, it aggregates the features of adjacent nodes and edges to generate the local neighborhood information surrounding the node. Subsequently, in the *COMBINE* step, this neighborhood information is integrated with the original features of node to iteratively refine the node embeddings. Formally, the message passing process can be defined as follows:

$$\alpha_u^l = \text{AGGREGATE}(\{h_u^{l-1} \mid u \in N(v)\}, \{h_{v,u}^{l-1}\}), \quad (1)$$

$$h_v^l = \text{ReLU}(\text{MLP}(\text{COMBINE}(h_v^{l-1}, \alpha_u^l))), \quad (2)$$

where α_u^l is the neighborhood aggregation information at layer l , h_v is the node embedding of v , $N(v)$ is the neighbor set of v , $h_{v,u}$ is the embedding of the edge between v and u . After L iterations, the i th molecular node-view graph embedding $z_{\tau,i}^N$ in task τ can be obtained by averaging or summing all node embeddings through the *READOUT* function:

$$z_{\tau,i}^N = \text{READOUT}(\{h_v^L : v \in V\}). \quad (3)$$

Subgraph-view encoder: Molecular property is closely related to the property of its substructure. In other words, the local substructure of the molecule plays an important role in the molecular property prediction. Therefore, this paper studies how to integrate local substructure information into molecular embedding. Inspired by NGNN [31], we regard the k -hop neighbors of node v as a local substructure, then the GNN model is applied to all substructures to extract the feature of substructure. The process is shown in Fig. 2(b).

First, each node in the original graph is regarded as the center node in turn and the k -hop neighbors of the center node are selected as a subgraph, and a set of subgraphs $\{g_1, g_2, \dots, g_n\}$

is obtained. Then, a GNN-based encoder is used to obtain the embeddings of the subgraphs. Finally, the i th molecular subgraph-view graph embedding $z_{\tau,i}^S$ can be obtained by aggregate all subgraph embeddings through the *READOUT* function.

To ensure that the obtained graph embedding contain as much information of the molecule as possible, the node-view graph embedding and the subgraph-view graph embedding are fused to form the final graph embedding $z_{\tau,i}^G$ by using (4):

$$z_{\tau,i}^G = \lambda_1 z_{\tau,i}^N + \lambda_2 z_{\tau,i}^S, \quad (4)$$

where λ_1, λ_2 is the weight parameters.

D. Relation Graph Learning Module

Labeled data is often limited in molecular property prediction, which raises the challenge for supervised learning. In fact, molecules with the same properties tend to share similar molecular structures [27]. For example, Aldehydes and ketones belong to a type of organic compound that contain a carbonyl (C = O) functional group. The carbonyl group play an important role in a set of common properties such as electrophilicity, reactivity and solubility. Accordingly, we can establish communication between molecules by capturing structural and functional similarities between molecules, allowing molecules to extract additional information from other molecules to augment their feature.

To utilized the relationship between molecules, PAR [26] builds a relation graph based on the distances between molecular representations. HSL-RG [27] utilizes graph kernels to establish the relation graph based on the similarity of molecular graph. In this paper, we also construct a relation graph by utilizing the similarity between molecules in a different way, on which molecules can communicate with each other, allowing limited

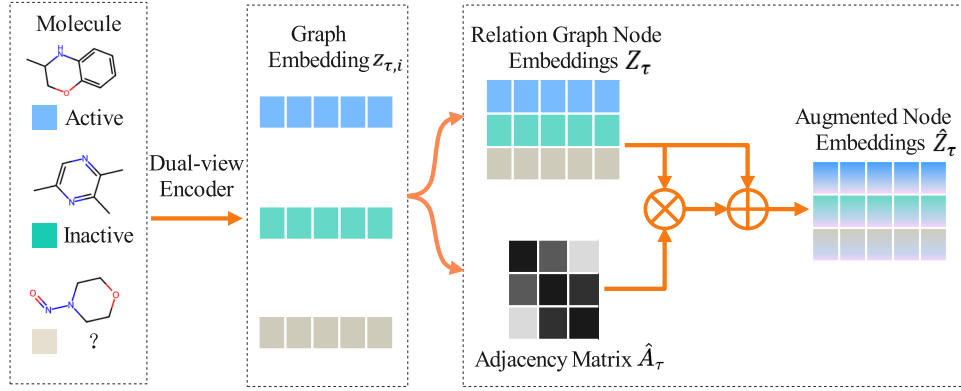


Fig. 3. Architecture of the relation graph learning module.

supervisory information to be transferred among molecules. The architecture is shown Fig. 3.

Specifically, a query molecule in query set Q_{τ} and $2K$ molecules in support set S_{τ} are taken as the nodes of relation graph. Let $Z_{\tau} \in R^{(2K+1,D)}$ denotes the all node embeddings of this relation graph where the i th row corresponds the i th molecular graph embedding $z_{\tau,i}$ obtained by dual-view encoder, which could be $z_{\tau,i}^N$, $z_{\tau,i}^G$, or $z_{\tau,i}^S$, D is the dimension of embedding. To capture the relationships between molecules, the cosine similarity is used to construct the relation graph adjacency matrix $A_{\tau} \in R^{(2K+1,2K+1)}$, where the $A_{\tau}(i,j)$ denotes the similarity between $z_{\tau,i}$ and $z_{\tau,j}$, which is calculated as:

$$A_{\tau}(i,j) = \frac{z_{\tau,i} \cdot z_{\tau,j}}{\|z_{\tau,i}\| \cdot \|z_{\tau,j}\|}. \quad (5)$$

To ensure communication between similar molecules while avoiding mutual interference between dissimilar molecules, we further update the adjacency matrix A_{τ} :

$$\hat{A}_{\tau}(i,j) = \begin{cases} A_{\tau}(i,j), & \text{if } A_{\tau}(i,j) \geq \sigma \\ 0, & \text{else} \end{cases} \quad (6)$$

where σ is a threshold. To minimize the loss of original information, a residual connection is introduced to connect the original embeddings and the updated embeddings, where the augmented relation graph node embeddings $\hat{Z}_{\tau}$ is obtained by communicate each node embedding with other node embeddings on this relation graph. Z_{τ} is updated as:

$$\hat{Z}_{\tau} = \text{LeakyReLU} \left(\beta_1 Z_{\tau} + \beta_2 \left(\hat{A}_{\tau} \cdot Z_{\tau} \cdot W_{\tau} \right) \right), \quad (7)$$

where W_{τ} is learnable parameter, β_1, β_2 is the weight parameters.

We denote $\hat{z}_{\tau,i}$ is the augmented molecular graph embedding corresponding to the i th element of $\hat{Z}_{\tau}$. Notably, $z_{\tau,i}^N$, $z_{\tau,i}^G$, and $z_{\tau,i}^S$ extracted by dual-view encoder are all participating in the above relation graph learning process, which obtains the $\hat{z}_{\tau,i}^N$, $\hat{z}_{\tau,i}^G$, and $\hat{z}_{\tau,i}^S$ respectively.

E. Prediction Module

To predict the probability of each property of the molecule, the augmented final graph embedding is fed into a downstream

MLP classifier. The probability can be calculated as: $\hat{y}_{\tau,i}^G = \text{MLP}(\hat{z}_{\tau,i}^G)$, where $\hat{z}_{\tau,i}^G$ denotes the i th augmented final graph embedding in task τ . The prediction loss is defined as the binary cross-entropy loss between the prediction $\hat{y}_{\tau,i}^G$ and the ground truth $y_{\tau,i}$:

$$L_{pred} = \sum_{(x_{\tau,i}, y_{\tau,i}) \in S_{\tau}} (y_{\tau,i} \log \hat{y}_{\tau,i}^G + (1 - y_{\tau,i}) \log (1 - \hat{y}_{\tau,i}^G)), \quad (8)$$

where S_{τ} denote the support set of the task τ . Further, to stabilize the training process of the dual-view encoder, the augmented graph embeddings $\hat{z}_{\tau,i}^N$ and $\hat{z}_{\tau,i}^S$ obtained from node-view encoder and subgraph-view encoder are also fed into the classifier to obtain the predictions $\hat{y}_{\tau,i}^N$ and $\hat{y}_{\tau,i}^S$ independently. The absolute loss function is introduced to enforce the predictions of the two classifiers to be close with each other, which is termed as disagreement loss:

$$L_{dis} = \sum_{(x_{\tau,i}, y_{\tau,i}) \in S_{\tau}} |\hat{y}_{\tau,i}^N - \hat{y}_{\tau,i}^S|. \quad (9)$$

The overall loss function is defined as the sum of prediction loss and disagreement loss:

$$L_{total} = L_{pred} + \rho L_{dis}, \quad (10)$$

where ρ is a weight hyper-parameter.

F. Property-Guided Feature Augmentation Meta-Learning

To ensure that our method could better adapt to the few-shot scenario, we adopt the MAML-based meta-learning framework [24]. During meta-training, meta-learner needs to learn a good initialization parameter θ from N_t tasks $\{T_{\tau}\}_{\tau=1}^{N_t}$. For each task T_{τ} , base-learner needs to use the support set loss $L_{T_{\tau}}^S$ to update the meta-parameter θ to task-specific parameter θ'_{τ} through gradient descent:

$$\theta'_{\tau} = \theta - \alpha \nabla_{\theta} L_{T_{\tau}}^S(\theta), \quad (11)$$

where α is a learnable parameter representing the step size of the gradient update, $L_{T_{\tau}}^S(\theta)$ is obtained by using (10). Subsequently, the base-learner uses the task-specific parameter θ'_{τ} to calculate

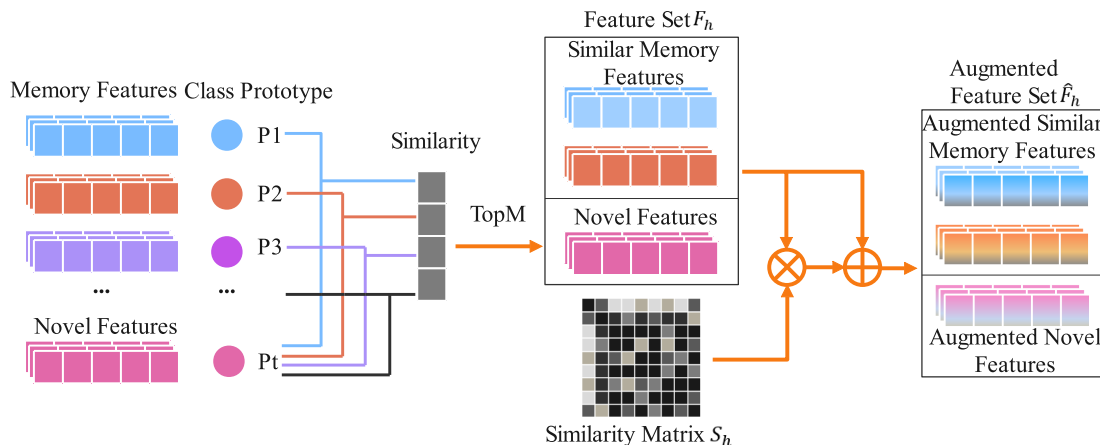


Fig. 4. Architecture of property-guided feature augmentation module.

the query set loss $L_{T_\tau}^Q(\theta'_\tau)$. Finally, meta-parameter is updated by integrating the query loss of all tasks:

$$\theta^* = \operatorname{argmin}_{\theta} \sum_{\tau=1}^{N_t} L_{T_\tau}^Q(\theta'_\tau). \quad (12)$$

During meta-testing, the meta-learner has learned a good initialization parameter and will be further fine-tuned on the support set in novel task, which contains K active molecules and K inactive molecules on the novel property. After fine-tuning, the model is able to predict the activity of the query molecule on the novel property.

However, for a novel property, the number of molecules are limited in the few-shot scenarios, which have limited information and their distribution cannot well represent the real distribution of the molecules with this property. Meta-learner is difficult to ensure that the parameters can be tuned in the optimal direction with a limited number of molecules. Consider that molecules with similar properties have similar molecular structures, if we know the similarity between the novel property and other properties, we can transfer additional information from the molecules with similar properties to the molecules with novel property. To this end, we design a property-guided feature augmentation module, as shown in Fig. 4.

Before feature augmentation, a series of molecular embeddings of each property are stored in the meta-training stage and treated as memory features. The molecular embeddings of novel property are extracted by the encoder as novel features. For each property, we calculate the mean of corresponding memory or novel features to obtain the class prototype P_τ by using (13), which can be used as the representation of current property:

$$P_\tau = \operatorname{Mean}(\{z_{\tau,i}\}_{i=1}^K), \quad (13)$$

where $z_{\tau,i}$ denote the i th memory or novel feature corresponding to current task τ . During meta-testing, we calculate the cosine similarity between the class prototype obtained by novel features and each class prototype obtained by memory features.

Algorithm 1: Meta-Training Procedure of PG-DERN.

Input: initial parameters θ

Output: trained parameters θ , memory features F_m

while not done **do**

sample a batch of task T_τ .

for all T_τ **do**

sample support set S_τ and query set Q_τ from T_τ .

obtain $z_{\tau,i}^N, z_{\tau,i}^G, z_{\tau,i}^S$ for each molecule by the encoder and augment them by (5)–(7).

store the memory features corresponding to current property.

calculate the training loss $L_{T_\tau}^S$ on S_τ by (10).

fine-tune θ to θ'_τ by (11).

calculate the testing loss $L_{T_\tau}^Q$ on Q_τ .

end for

update θ by (12).

end while

Then, we select M properties with the highest similarity scores, and combine their memory features and the novel features as a feature set F_h . A similarity matrix S_h is constructed by calculating the cosine similarity between each feature of F_h . According this, we can further augment the novel features by transfer the information from the similar memory features to the novel features. As the similarity between features is calculated, the transfer of irrelevant information is reduced during feature augmentation by assigning lower weights to the features that are less similar to the novel features. The F_h is first updated as:

$$\hat{F}_h = \operatorname{LeakyReLU}(\varphi_1 F_h + \varphi_2 (S_h \cdot F_h \cdot W_h)). \quad (14)$$

Then, the augmented novel features can be obtained from $\hat{F}_h$. As before, feature augmentation should be applied to the three embeddings of each molecule extracted by dual-view encoder in the fine-tuning stage. After feature augmentation, meta-learner fine-tunes the meta-parameter on the support set with the augmented novel features and is evaluated on the query set.

TABLE I
INFORMATION OF FOUR BENCHMARK DATASETS

Dataset	Tox21	SIDER	MUV	ToxCast
Attribute	Toxicity	Side effect	Activity	Toxicity
Compounds	7831	1427	93127	8615
Training Tasks	9	21	12	450
Testing Tasks	3	6	5	167

The meta-training algorithm of PG-DERN is shown in Algorithm 1.

III. EXPERIMENTS

In this section, we conducted extensive experiments on the benchmark datasets and demonstrate the advantages of PG-DERN with compared to other baseline models and methods. In addition, we verified the effective of each module and the practicality of PG-DERN on molecular property prediction, and provide the relevant analysis.

A. Benchmark Dataset

Four benchmark datasets are used to validate our method for molecular property prediction, which are Tox21 [32], SIDER [33], MUV [34], and ToxCast [35].

Tox21: The Tox21 contains 7831 compounds and 12 tasks, where each task represents the toxicological activity performance of the compounds against different targets. For example, the task “NR-AR” contains the endocrine disrupting chemicals that interact with nuclear receptor (NR), including androgen receptor (AR). **SIDER:** The SIDER contains 1427 compounds and 27 tasks corresponding to 27 body organs, where each task represents an adverse reaction that a marketed drug may cause to an organ. **MUV:** The MUV contains 93127 compounds and 17 tasks, where each task represents a different biological activity. **ToxCast:** The ToxCast contains 8615 compounds and 617 tasks, where each task represents a different biological mechanism and toxicity correlation.

We used the same approach to divide the task into training and testing tasks as in PAR [26]. For example, 9 tasks in Tox21 are used as training tasks and the remaining 3 tasks are used as testing tasks. Each task is a binary classification task, indicating whether a molecule has a certain property. The information for each dataset is shown in Table I.

The features of the molecule include atom type, atom chirality, bond type and bond direction for atoms and chemical bonds. This set of features formed the initial set of atom and bond features, which are fed into GIN layers with a molecular graph adjacency matrix to obtain the molecular embedding.

B. Baseline Methods

We categorized the following benchmark methods into three different categories and compared them with our PG-DERN. The first category is classical few-shot learning method including Siamese [36], ProtoNet [22], TPN [37], EGNN [38],

IterRefLSTM [25]. The second category is MAML-based meta-learning method without pre-trained molecular encoders, including MAML [24], MAML+MetaMix [39], Sharp-MAML [40], MAML+MTA [18]. The third one is MAML-based method with pre-trained molecular encoders, including Pre-GNN [41], Meta-MGNN [9], Pre-PAR [26], HSL-RG [27].

C. Model Reproduction

In the dual-view encoder module, the node-view encoder is a 5-layer pre-trained GIN of Pre-GNN [41] and the subgraph-view encoder is a 2-layer GNN, the weight of their graph embedding is 0.9 and 0.1 separately. We use a 2-layer MLP as the classifier to predict the property of molecule. The size of molecular embedding is 300 and the size of hidden layer is set to 128, and the weight of the disagreement loss is set to 0.01. In the relation graph learning module, the weight of the original embedding is set to 0.1 and the weight of the updated embedding is set to 0.9 in the final residual connection layer. In the property-guided feature augmentation module, the number of similar properties N^P is 3, the residual layer is set in the same way as the relation graph learning module. Other hyper-parameters follow the settings of Pre-PAR [26].

Pytorch is used to implement the model on Nvidia GeForce RTX 4090. The average of the meta-training time of the model on four datasets for each epoch is 0.043 minutes, after 5000 epochs, the entire training process can be completed in about 4 hours. The memory consumption is 3222 MiB in the training process. The inference time indicating the time for predicting property of a single molecule based on the trained model, which is 0.01 s on average.

D. Effects of PG-DERN and Other Baseline Methods

The ROC_AUC score is used as the evaluation metric for comparison. For each experiment, we ran five times using different random seeds to obtain reliable and robust model results. We performed the tests in the 10-shot and 1-shot settings respectively, and reported the mean and standard deviation of the ROC_AUC scores for all meta-test tasks.

The performance of all the methods as shown in Table II. The results of baseline methods are acquired from published results [26], [27]. Across the four datasets, PG-DERN outperforms the previous best baseline (HSL-RG) on average and outperforms HSL-RG in six out of eight results, with an average improvement of 0.3% in each dataset compared to the HSL-RG.

According to the statistics, the MAML-based meta-learning methods show better performance than traditional few-shot learning methods in current molecular property prediction task. This is because MAML could optimize the generalization performance of the model on different tasks, rather than achieving the best performance on a specific task, which makes the MAML more robust in the face of novel tasks.

In addition, the methods with pre-trained encoder (e.g., PG-DERN, HSL-RG, Pre-PAR, Meta-MGNN, Pre-GNN) presents a superior performance compared to other methods that learned from scratch. It is due to the pre-trained methods use a large amount of molecular data, allowing the GNN to get better

TABLE II
ROC_AUC SCORES ON FOUR BENCHMARK DATASETS OF MOLECULAR PROPERTY PREDICTION

Method	Tox21		SIDER		MUV		ToxCast	
	10-shot	1-shot	10-shot	1-shot	10-shot	1-shot	10-shot	1-shot
Siamese	80.40±0.35	65.00±1.58	71.10±4.32	51.43±3.31	59.96±5.13	50.00±0.17	-	-
ProtoNet	74.98±0.32	65.58±1.72	64.54±0.89	57.50±2.34	65.88±4.11	58.31±3.18	63.70±1.26	56.36±1.54
TPN	76.05±0.24	60.16±1.18	67.84±0.95	62.90±1.38	65.22±5.82	50.00±0.51	62.74±1.45	50.01±0.05
EGNN	81.21±0.16	79.44±0.22	72.87±0.73	70.79±0.95	65.20±2.08	62.18±1.76	63.65±1.57	61.02±1.94
IterRefLSTM	81.10±0.17	80.97±0.10	69.63±0.31	71.73±0.14	49.56±5.12	48.54±3.12	-	-
MAML	80.21±0.24	75.74±0.48	70.43±0.76	67.81±1.12	63.90±2.28	60.51±3.12	66.79±0.85	65.97±5.04
MAML+MetaMix	81.83±0.15	81.13±0.21	73.34±0.18	72.69±0.31	67.38±1.03	66.39±1.22	70.22±0.92	70.03±1.14
Sharp-MAML	75.37±0.23	74.59±0.56	71.02±0.81	68.43±0.96	65.52±2.01	65.12±2.98	67.56±1.01	66.49±1.98
MAML+MTA	83.33±0.34	83.09±0.42	77.43±0.23	74.63±0.37	69.70±1.05	67.42±1.12	73.31±0.89	71.58±1.26
Pre-GNN	82.14±0.08	81.68±0.09	73.96±0.08	73.24±0.12	67.14±1.58	64.51±1.45	73.68±0.74	72.90±0.84
Meta-MGNN	82.97±0.10	82.13±0.13	75.43±0.21	73.36±0.32	68.99±1.84	65.54±2.13	-	-
Pre-PAR	84.93±0.11	83.01±0.09	78.08±0.16	74.46±0.29	69.96±1.37	66.94±1.12	75.12±0.84	73.63±1.00
HSL-RG	85.56±0.28	84.09±0.20	78.99±0.33	77.53±0.41	71.26±1.08	68.76±1.05	76.00±0.81	74.40±0.82
PG-DERN	85.25±0.29	84.12±0.08	79.62±0.32	77.69±0.38	71.65±0.26	69.66±1.02	75.21±0.19	74.51±0.17

The bold values represent the optimal result among all methods.

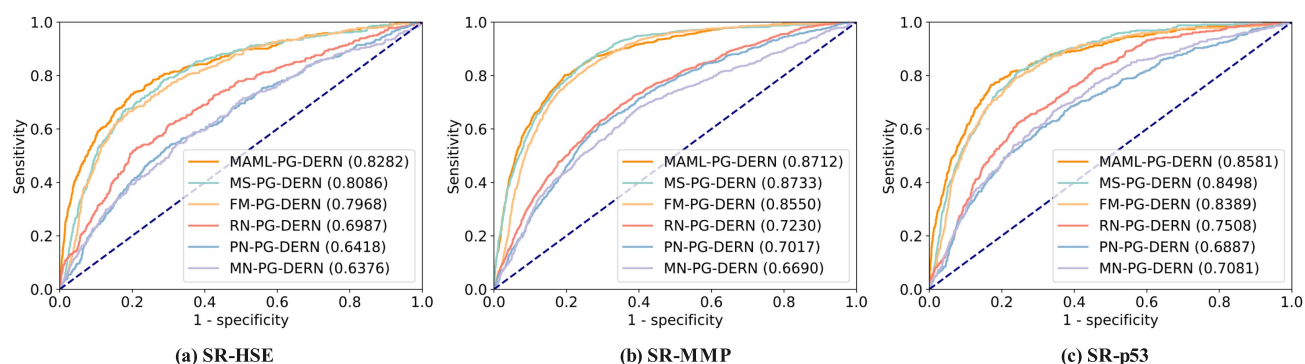


Fig. 5. ROC_AUC scores of different meta-learning methods.

parameter initialization. We also observe that PG-DERN and Pre-PAR achieves better performance than Pre-GNN and Meta-MGNN, which could be attributed to their effective utilization of the relationship between molecules. PG-DERN captures the relationship between molecules by constructing a relation graph, which facilitates the propagation of information between molecules. Besides that, we believe that PG-DERN could refine the optimization process of meta-learning. By transferring the knowledge from the molecules with similar properties, PG-DERN augments the features of the molecules with novel property, which could guide the meta-learning parameters to be tuned in an optimal direction.

E. Effects of Applying Different Meta-Learning Strategies to the PG-DERN

The MAML is used as the meta-learning strategy in PG-DERN. To compare the effects of applying different meta-learning strategies to PG-DERN, five other meta-learning strategies are selected which are Meta-SGD (MS), FOMAML (FM),

Relation Network (RN), Prototypical Network (PN), and Matching Network (MN). Fig. 5 shows the ROC_AUC curves of the different meta-learning algorithms on the three test tasks of Tox21 and Table III exhibits the average ROC_AUC scores of different meta-learning strategies on the three tasks. MAML-PG-DERN demonstrates superior performance compared to the other methods, with MS-PG-DERN and FM-PG-DERN following behind. In general, the computational complexity of MAML is higher than that of Meta-SGD and FOMAML due to that MAML requires multiple gradient updates to the entire model during training. The Meta-SGD does not involve multiple parameters for the entire model. FOMAML is a variant of MAML, which uses only first-order derivative information, thus simplifying the computation. To sum up, the MAML and its variants outperform other meta-learning strategies by using PG-DERN.

F. Ablation Study

1) *Ablation Study for Dual-View Encoder Module:* To verify the effectiveness of dual-view encoder, we designed two ablation

TABLE III
AVERAGE ROC_AUC SCORES OF DIFFERENT META-LEARNING METHODS ON TOX21

	MAML-PG-DERN	MS-PG-DERN	FM-PG-DERN	RN-PG-DERN	PN-PG-DERN	MN-PG-DERN
Average	85.25	84.39	83.02	72.42	67.74	67.16

The bold values represent the optimal result among all methods.

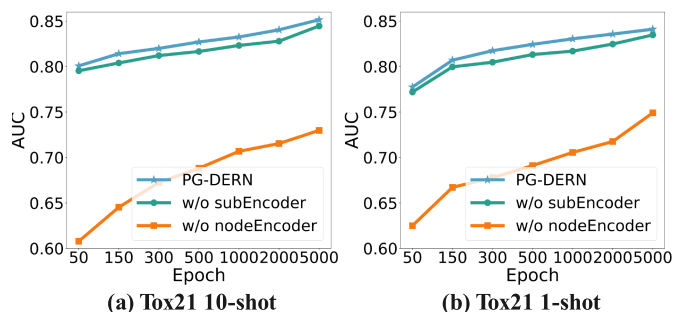


Fig. 6. Ablation experiments for dual-view encoder module on Tox21.

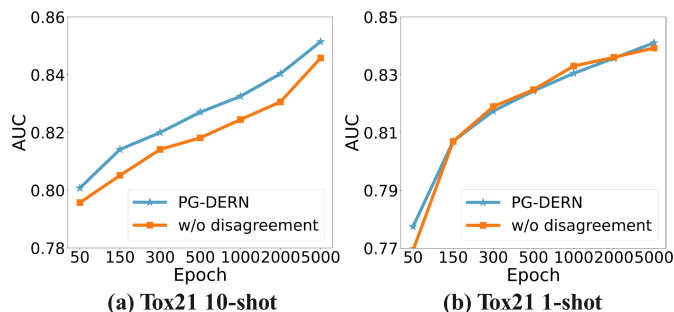


Fig. 8. Ablation experiments for prediction module on Tox21.

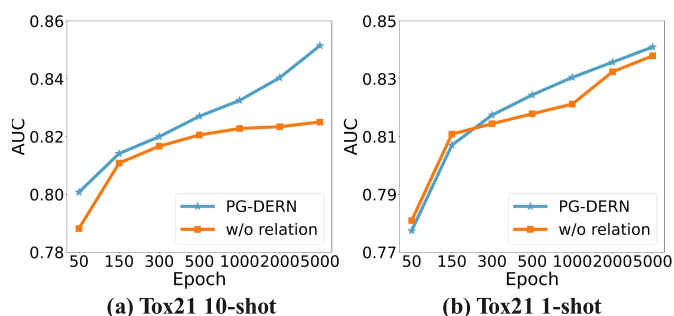


Fig. 7. Ablation experiments for relation graph learning module on Tox21.

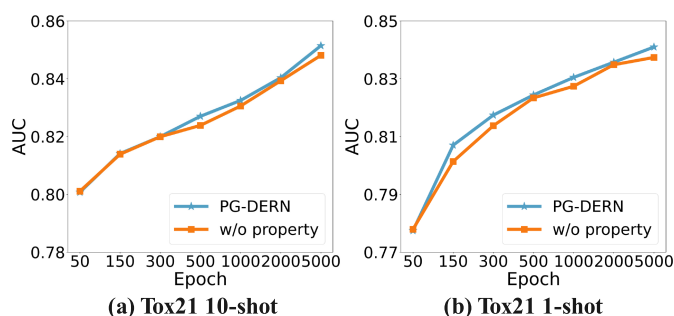


Fig. 9. Ablation experiments for property-guided feature augmentation module on Tox21.

models: w/o subEncoder and w/o nodeEncoder. Among them, w/o subEncoder is the PG-DERN without the subgraph-view encoder, and w/o nodeEncoder is the PG-DERN without the node-view encoder. Their ROC_AUC scores with respect to different epochs are shown in Fig. 6. It is observed that the performance of w/o subEncoder and w/o nodeEncoder are consistently lower than PG-DERN. From our perspective, this is attributed to the fusion of the molecular representation obtained by the two encoders, which complement each other with the molecular feature information, enabling the molecular representation to own a stronger expressive capability. w/o nodeEncoder is significantly lower than the other models, it is possible that the node-view encoder uses pre-trained GIN and already has a good generalization ability.

2) Ablation Study for Relation Graph Learning Module: A further experiment was conducted to demonstrate the impact of the relation graph learning module on the performance of model. w/o relation is a ablation model that removes the relation graph learning module. We conducted a comparative analysis with the PG-DERN, and the experimental results are depicted in Fig. 7. It is evident that the ablation model w/o relation exhibits lower performance compared to the PG-DERN. We consider that the relation graph learning module is effective in capturing the relationship between molecules. Meanwhile,

knowledge communication between molecules can effectively enhance the feature representation of the molecules, making it easier for the model to learn and understand the patterns of the data. It further draws an important conclusion that utilizing the relationship between molecules is beneficial in improving the performance of the model.

3) Ablation Study for Prediction Module: To evaluate the disagreement loss in the prediction module, an ablation model removing disagreement loss, w/o disagreement, was added for comparison in the revision. As shown in Fig. 8, the performance of w/o disagreement is lower than that of PG-DERN. It indicates that the disagreement loss plays a positive role in enhancing the prediction of the model. We believe that the disagreement loss allows both encoders to converge at similar rates as a way to maintain a steady state for dual-view encoder training, thus improving the robustness of model.

4) Ablation Study for Property-Guided Feature Augmentation Module: To illustrate whether the property-guided feature augmentation module is valid or not, an ablation model was designed without it, named w/o property. Fig. 9 shows the experimental results of the PG-DERN and the w/o property, which indicates that the w/o property consistently underperforms in the prediction. It further validates our hypothesis that by rationally fusing molecular information with similar properties, the

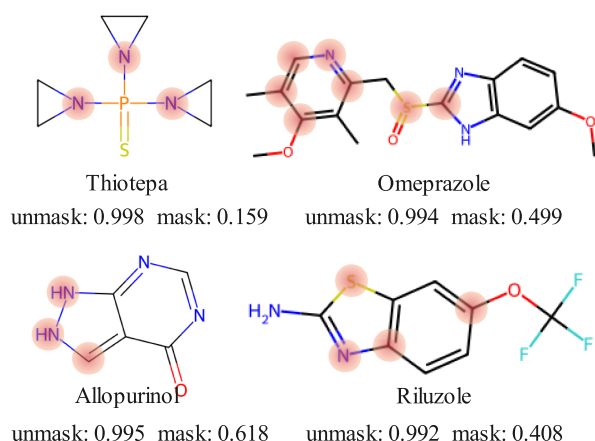


Fig. 10. Analysis of the interpretability of the PG-DERN.

molecular representation can be effectively augmented, which can better guide the tuning of meta-parameters and achieve more accurate prediction.

G. Analysis of the Interpretability of the PG-DERN

An experiment was conducted to improve the interpretability of the PG-DERN and explore the relationship between these features and molecular organ toxicity, in which highly contributing atoms are highlighted by selective node masking. Specifically, the difference between the prediction scores of masked and unmasked nodes is treated as the contribution of atoms. Fig. 10 illustrates the four molecules with hepatotoxicity in SIDER dataset, in which the top 30% of atoms with high contribution scores are highlighted. The prediction results of masking and unmasking significant atoms are exhibited under each molecule.

It is observed from the experiment that masking the important atoms significantly affect the performance of the model, suggesting that these atoms or functional group may be associated with hepatotoxicity. For example, Pyrazolidine-containing drugs are known to cause glutathione depletion and promote hepatotoxicity [42]. The highly contributing atoms in Allopurinol belong to Pyrazolidine group, which may be a potential mechanism of Allopurinol causes hepatotoxicity. In addition, the liver injury caused by Riluzole may be related to the structure of Thiazole, which may produce toxic metabolites during biotransformation [43].

Overall, the feature of atoms with high contributions could be extracted through the PG-DERN and play an important role in prediction. Moreover, it is helpful in understanding the relationship between the molecule and their specific properties by PG-DERN, which is significant in the development of novel drug molecules.

H. Analysis of the Classification Ability of PG-DERN

To intuitively display the classification capability of model, a visualization experiments was performed on the SR-MMP task on the Tox21 dataset. The t-SNE technology was applied to map the molecular embeddings generated by PG-DERN and

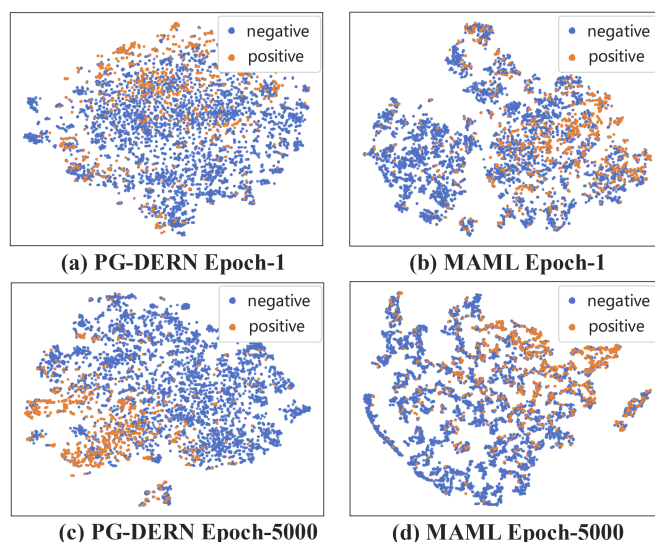


Fig. 11. Distribution of molecular embeddings generated by PG-DERN and MAML on Tox21.

the baseline model MAML onto a two-dimensional space, as shown in Fig. 11. Fig. 11(a) and (b) show the distribution of embeddings at the first epoch of training, while Fig. 11(c) and (d) show the distribution of embeddings at the last epoch of training. The orange dots represent molecules that are active in SR_MMP property (positive samples), while the blue dots represent molecules that are inactive in SR_MMP property (negative samples). From a Fig. 11(a) and (c), the orange dots and the blue dots are gradually separated, which indicates that our training of the model is meaningful. From a horizontal perspective, in Fig. 11(c), the orange dots are mainly concentrated in the lower left corner, while the blue dots are mainly concentrated in the upper right corner. However, a lot of blue and orange dots in Fig. 11(d) are mixed together, which also demonstrates the superiority of the proposed model.

I. Analysis of the Role of Feature Augmentation

The SR_MMP property in the Tox21 dataset was leveraged to explain the significance of the property-guided feature augmentation module. 5 specific molecules were selected from the 918 molecules with SR_MMP properties as the known data and the remaining molecules as the unknown data. We will consider a practical issue and try to answer it: Assuming that the distribution of the known data deviates from the actual distribution of this class of molecules, how can we better predict the unknown data based on the known data?

In the experiment, we first extracted the embedding of this set of molecules by the encoder, and then projected them into the 2D space by t-SNE, as shown in Fig. 12(a), where the orange points represent the known data and the blue points represent the unknown data with same property. The distribution of the blue point could be regarded as the actual distribution. Next, the feature representations of the known data were augmented by the property-guided feature augmentation module. Through the experiment, we have learned that NR_AHR, NE_ER, SR_ARE

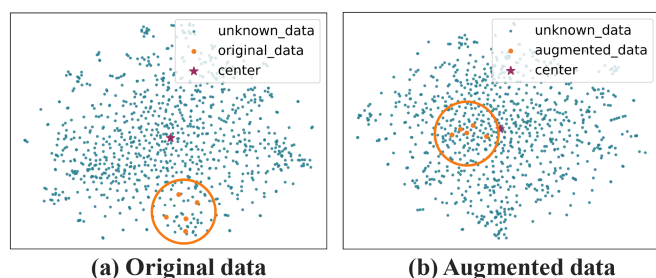


Fig. 12. Distribution of molecular embeddings before and after molecular feature augmentation.

TABLE IV
EVIDENCES OF TOP TEN PREDICTED DRUGS

Drug	Evidence
Esomeprazole	Drugbank DB00736
Allopurinol	SIDER
Omeprazole	Drugbank DB00338
Lansoprazole	PMID: 33625647
Dexlansoprazole	Drugbank DB05351
Pentostatin	PMID: 10789823
Rabeprazole	PMID: 33625647
Dapiprazole	None
Dacarbazine	None
Pantoprazole	PMID: 33625647

are similar to SR_MMP. Following this, a set of molecules were extracted from each of these properties and transfer their knowledge to the original known data through the property-guided feature augmentation module. Finally, we re-projected the unknown data and the augmented known data into the 2D space as shown in Fig. 12(b). We observe that the known data are closer to the center of the class. It is possible that the known data incorporate structural knowledge of more similar molecules, adjusting their distributions to a certain extent. Following this, the augmented molecular features can be used to better fine-tune the meta-parameters and predict unknown data.

J. Case Study for the Prediction of Adverse Cardiac Effect of Drug by PG-DERN

During the clinical trial phase of drug discovery, identifying adverse drug reactions to human organs is helpful in determining the safety and tolerability of a drug, thereby reducing the risks of the drug. To validate the clinical utility of PG-DERN, we used PG-DERN to predict potential drugs in the Drugbank dataset [44] that may have adverse cardiac effects. To implement meta-learning, ten compounds known to produce adverse reactions to the heart were collected to construct a support set to fine-tune the model parameters.

According to prediction results of PG-DERN, the top ten drugs in terms of prediction score are listed in Table IV. Eight of these drugs have been validated for possible adverse cardiac effects. It is noteworthy that six of these eight drugs belong to

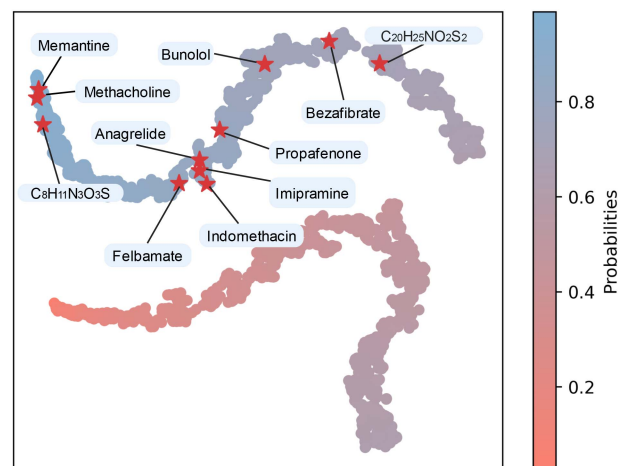


Fig. 13. Distribution visualization of adverse cardiac effect drugs from Drugbank inferred by PG-DERN.

the class of proton pump inhibitor (PPI), which are Esomeprazole, Omeprazole, Lansoprazole, Dexlansoprazole, Rabeprazole, and Pantoprazole. In general, PPIs may inhibit the activity of dimethylarginine dimethylamino hydrolase (DDAH), an enzyme critical to cardiovascular health, leading to an increased risk of cardiac or cardiovascular events.

Moreover, The PG-DERN only is updated by one epoch on ten molecules in the fine-tuning process, which takes about 5 seconds, and the 1704 molecules in the Drugbank dataset was predicted by the PG-DERN, which takes only approximately 0.27 minutes. For each molecule, it only takes no more than 0.02 s to obtain its results. It is demonstrated that the model can be efficiently applied to the online molecular property prediction tasks.

In addition, the probability of each drug being inferred to potentially cause adverse cardiac effects is calculated, and their overall distributions are visualized using T-SNE, as shown in Fig. 13. Based on the predictions of the model, drugs that may cause adverse cardiac effects are easily separated from other drugs into two categories. Further, 12 drugs known to potentially cause adverse cardiac effects are used as a validation set and predicted using PG-DERN, of which 11 are visualized into a high probability space as shown in Fig. 13. It demonstrates the powerful generalization ability of PG-DERN in drug discovery.

IV. CONCLUSION

In this paper, we propose a few-shot learning model for molecular property prediction, named PG-DERN. Three novel modules are introduced in the PG-DERN to improve the performance. The dual-view encoder module could capture molecular graph structure, atom and chemical bond features to enhance the comprehensiveness of molecular embedding. The relation graph learning module constructs a molecular relation graph by capturing the similarities between molecules, allowing the molecules to communicate and transfer information efficiently. In addition, the property-guided feature augmentation module is proposed to transfer knowledge from molecules with similar

property to the molecules with novel property by establishing correlations between properties. Through knowledge transfer, meta-learning is able to efficiently fine-tune meta-parameters for the new few-shot tasks. We conducted comparative experiments on four public datasets. A series of experimental results demonstrate the superiority of the PG-DERN and validate the effectiveness of proposed modules.

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